Status	Finished
Started	Wednesday, 28 October 2026, 4:57 PM
Completed	Wednesday, 28 October 2026, 5:03 PM
Duration	6 mins 31 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **2**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sort()

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. All of the above
c. Searching
d. Selection

The correct answer is: All of the above

Question **6**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates
b. It complicates the analysis process
c. It has no effect on the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Merge Sort
- c. Insertion Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes the data unsorted
- b. It slows down the process
- c. It complicates the selection of items
- d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Divide and conquer
- c. Backtracking
- d. Greedy method

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection
b. Distribution
c. Extraction
d. Insertion

The correct answer is:
Insertion

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Dividing the input into smaller subproblems
c. Sorting the entire list sequentially
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It sorts data out of the box efficiently
c. It is less efficient than custom sorting algorithms
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It makes the search operation slower
- c. It increases the number of comparisons needed
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **15**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Sorting data in a non-sequential manner
- c. Using the divide-and-conquer approach
- d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass